

Software Requirements Specification (SRS)

SIM (Security Incident Mapping)

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1. Introduction

1.1 Purpose

The purpose of this document is to describe the software requirements for the Security Incident Mapping (SIM) system. The system is designed to provide a simple web-based platform through which members of the campus community can report security incidents quickly and efficiently. The document serves as a guide for the design, development, and evaluation of the system.

1.2 Scope

Security Incident Mapping (SIM) is a prototype web application developed to improve the reporting of security incidents within and around the campus environment. The system enables students, staff, and landlords to report incidents such as theft, assault, harassment, robbery, and other security-related concerns. The reports generated through the system are intended to support quicker awareness and response to campus security issues.

1.3 Intended Audience

This document is intended for:

Students

Staff

Landlords

Security Personnel

Course Lecturer

Software Developers

2. Overall Description

2.1 Product Perspective

SIM is a standalone web-based application that can be accessed using any modern web browser. The prototype was developed using HTML, CSS, and JavaScript and demonstrates the basic functionality required for reporting security incidents.

2.2 User Classes

Student

Students can submit reports about security incidents and choose whether to report anonymously.

Staff

Staff members can report incidents observed within the campus community.

Landlord

Landlords can report incidents occurring around their hostels or rental properties occupied by students.

Admin

The administrator manages submitted reports and monitors incident information.

3. Requirement Gathering Summary

3.1 Data Collection Method

The requirements for the SIM (Security Incident Mapping) system were gathered using a Google Forms questionnaire distributed to members of the campus community, including students, staff, and landlords. The survey was designed to understand the nature of security challenges on campus, current reporting methods, and users' expectations for a web-based reporting platform.

3.2 Summary of Findings



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Questions

Responses

6

Settings

Summary

Question

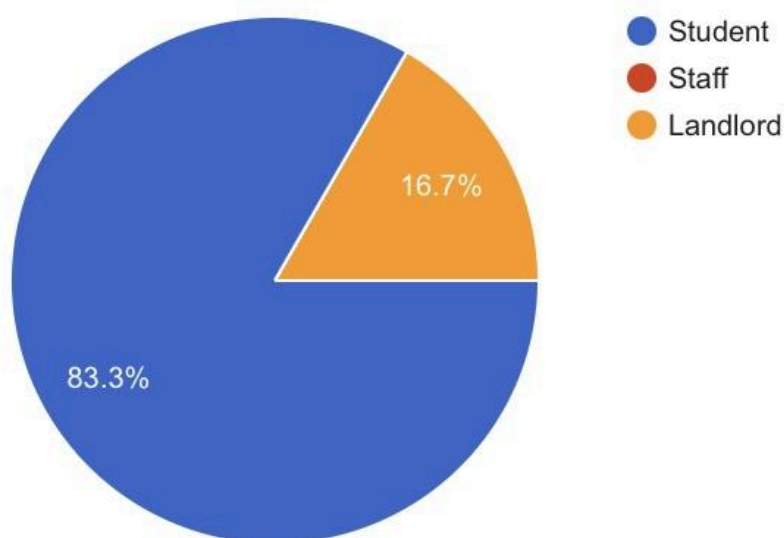
Individual

What is your role in the campus community



Copy chart

6 responses



Have you ever witnessed or



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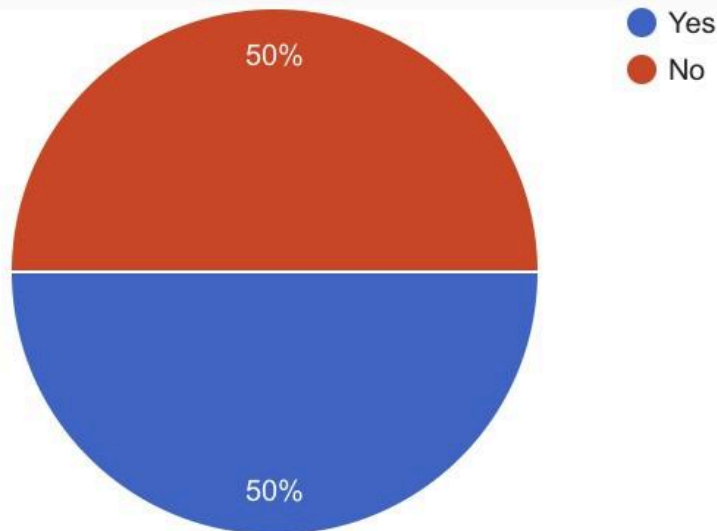
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Questions

Responses

6

Settings

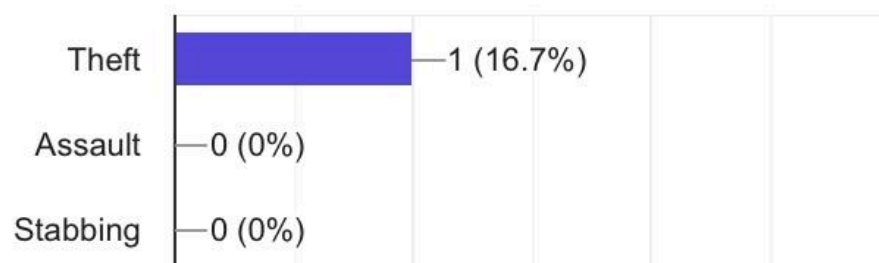


If yes, what type of incident was it?



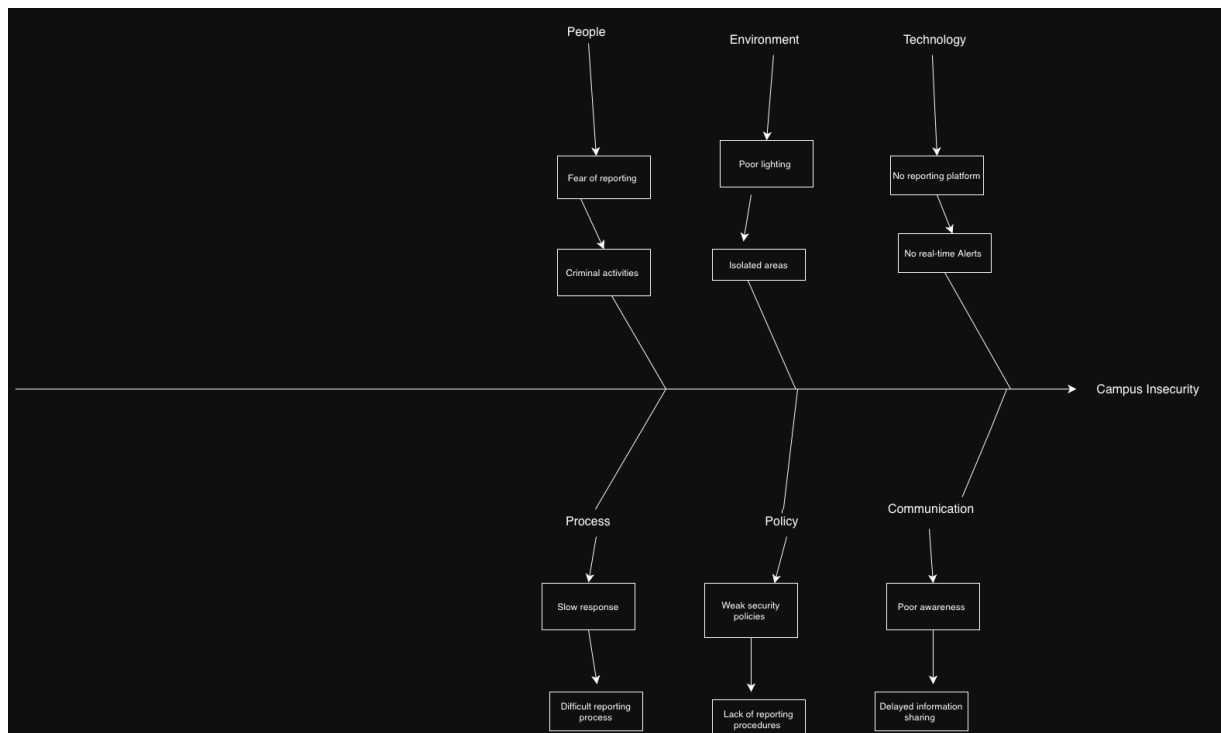
Copy chart

6 responses



The survey indicated that security incidents such as theft, harassment, assault, and robbery are common concerns within the campus environment. Many respondents stated that incidents are either not reported or are reported late due to fear, lack of awareness, or the absence of a simple reporting system.

Most respondents also indicated that they would be willing to use a web-based application that allows quick and anonymous reporting of security incidents. Features such as anonymous reporting, photo upload, emergency alerts, and report status tracking were identified as highly desirable.



The Ishikawa (Fishbone) Diagram identifies the major causes of campus insecurity. The causes are grouped into six categories: People, Environment, Technology, Process, Policy, and Communication. This analysis helped define the requirements of the proposed SIM system.

4. Functional Requirements

The SIM system shall provide the following functions:

FR1: The system shall allow users to submit a security incident report.

FR2: The system shall allow users to select the type of incident.

FR3: The system shall allow users to enter the location of the incident.

FR4: The system shall allow users to provide a description of the incident.

FR5: The system shall allow users to submit reports anonymously.

FR6: The system shall generate a unique reference number after a report is submitted.

FR7: The system shall display a confirmation message after successful submission.

FR8: The administrator shall be able to view submitted reports (future enhancement).

5. Non-Functional Requirements

Usability

The system shall provide a simple and user-friendly interface that can be used without technical knowledge.

Performance

The system shall display confirmation immediately after a report is submitted.

Availability

The system shall be accessible through modern web browsers on both mobile phones and computers.

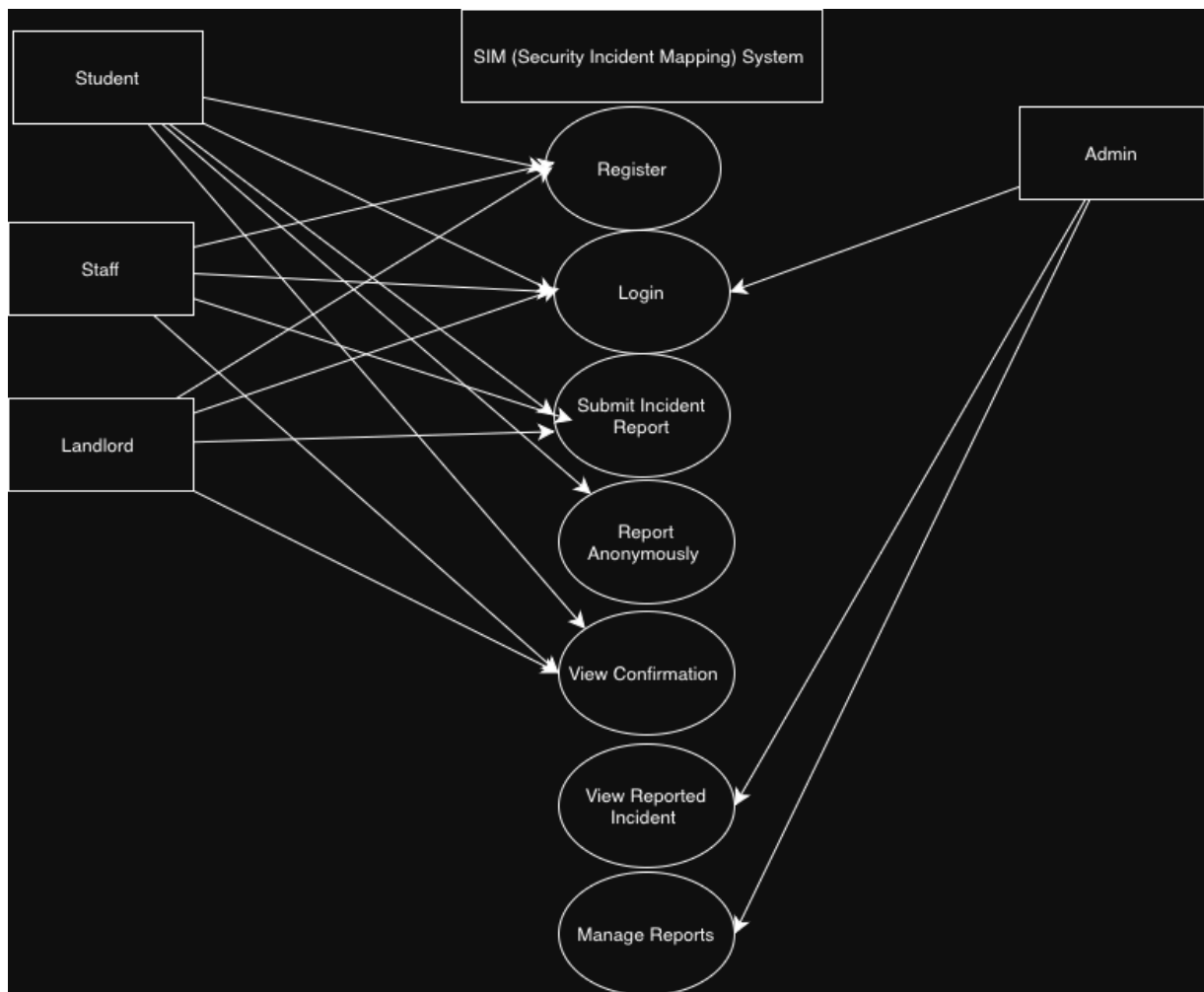
Security

Anonymous reports shall not require personal identification.

Reliability

The system shall provide consistent operation during normal usage.

6. Use Case Diagram



Explanation of the Use Case Diagram

The Use Case Diagram illustrates the interaction between the users and the SIM system.

Student

A student can register, log in, submit an incident report, report anonymously, and view the confirmation after submission.

Staff

A staff member can register, log in, submit an incident report, and view the confirmation message.

Landlord

A landlord can register, log in, submit an incident report, and view the confirmation message.

Administrator

The administrator logs into the system to view submitted reports and manage reported incidents.

7. Object Listing

Object Name	Attributes	Description
User	User ID, Name,Role	Represent a user of the system(Student, Staff,Landlord, or Admin).
Incident Report	Incident Type, location,Description, Date,Anonymous Status	Stores information about a reported security incident.
Confirmation	Reference Number, Submission Date	Confirms that an incident report has been successfully submitted.
Administrator	Admin ID, Username,Password	Manages reports submitted through the system.

8. System Architecture

The SIM system follows a simple client-side architecture suitable for a prototype.

Frontend

The user interface was developed using:

HTML

CSS

JavaScript

User Interaction

Users access the application using a web browser on a mobile phone or computer. They complete the incident report form and submit it through the interface.

Processing

JavaScript processes the user's input, generates a unique reference number, and displays a confirmation message after successful submission.

The proposed SIM system offers several benefits:

Fast reporting of security incidents.

Anonymous reporting to protect users.

Easy accessibility through a web browser.

Improved communication between the campus community and security personnel.

Better documentation of reported incidents.

Increased awareness of security issues.

Supports faster response to emergencies.

10. Limitations of the Prototype

The current prototype has the following limitations:

Reports are not stored in a database.

No user authentication.

No real-time notifications.

No GPS location tracking.

No administrator login interface.

No integration with campus security services.

These features can be added in future versions of the system.

11. Conclusion

The Security Incident Mapping (SIM) system was developed as a prototype to improve the reporting of security incidents within the campus environment. The system provides a simple and user-friendly interface through which students, staff, and landlords can report incidents quickly and conveniently.

The project demonstrates how web technologies such as HTML, CSS, and JavaScript can be used to address real-world security challenges. Although the current version is a prototype, it provides a strong foundation for future enhancements, including database integration, real-time notifications, user authentication, and administrative management.

Overall, the SIM system has the potential to improve campus safety by encouraging timely incident reporting and supporting better communication between members of the campus community and security authorities.